Employee-treatment language in technology acquisitions

A disclosure-first study of 133 transactions, with an unsupervised model of what their filings say about people

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Status: provisional. One human quality check was not run for this cycle. Section 9 says exactly what that means. Nothing in this report is presented as a validated finding.

1. What this study asks

- When one company buys another, **what does it put in writing about the people it is acquiring?**
- Not what it says in interviews. Not what happened afterwards. **What the binding documents disclose.**
- This is empirical corporate-finance and human-capital research. It is not stock prediction and not legal advice.

1.1 Why the question is answerable at all

- United States securities law forces buyers and targets to file certain documents publicly.
- Merger agreements routinely contain an **employee-matters article**: promises about pay, benefits, equity and severance after closing.
- Merger proxies contain sections on **what executives and directors receive** because of the deal.
- So there is a real, large, free corpus of text about how transactions treat employees.

1.2 What the question is not

- We are **not** measuring whether employees stayed. That requires employment data we do not have.
- We are **not** measuring whether employees were treated well. A promise in a contract is a promise.
- We are **not** claiming these deals represent all acquisitions. Section 9 explains why they cannot.

2. The prior approach, and exactly how it failed

The earlier version of this project selected deals by asking **"is the target an AI company?"** and then went looking for filings. This is the natural order and it does not work.

Step in the old approach	Result
AI keyword screen over company names and descriptions	119 candidate deals
Acquirer could not be found on the SEC system	**57 of 119 dropped**

Step in the old approach	Result
Machine-qualified, pending human review	35
Deals that produced any employee text at all	**13**
Unique employee passages	**72**
Share of that text from a single deal	**44%**

2.1 The diagnosis

- The acquirers that buy AI startups are disproportionately **private companies, private-equity funds, and foreign corporations**.
- Examples that could not be resolved: Progressive AE, Tie Industrial, LFM Capital, Siemens AG, AXA SA, BAE.
- **Entities like these do not file with the SEC.** There is no document to read, so there is no evidence, no matter how interesting the deal is.
- The screen also produced false positives, pulling in robotics and lawn-mower manufacturers on the strength of the word "autonomous".

2.2 The insight that changed the design

- Whether employee text exists is **almost entirely a property of the buyer, not the target.**
- The predictor is: **is the acquirer an SEC registrant that filed the transaction agreement?**
- Evidence from our own pilot, holding the target type constant:

Deal	Target	Employee passages
Intuit / Mailchimp	private	**84**
Fastly / Glitch	private	**0**
Oracle / Cerner	public	**541**
Take-Two / Zynga	public	**607**

- Both Mailchimp and Glitch were private targets. The difference is entirely on the buyer's side.
- **72 passages cannot support a topic model.** A half-sample stability test on the old corpus scored **0.246**. The result was not weak, it was absent.

3. The new methodology

3.1 The principle

- **Select on disclosure, not on subject matter.**
- Find transactions whose filings provably exist, retrieve them, and only then ask which ones are about AI.
- "AI" becomes a **label applied after selection**, which converts an unanswerable question into an answerable one: *among deals whose employee terms are public, how many are described in AI terms?*

3.2 Stage 1 — Build a candidate pool (offline, deterministic)

Rule frozen in `config/disclosure_pool.toml`, version `disclosure-first-v1`:

- The acquirer must resolve to an SEC filer identifier at **high or medium** confidence.
- The target's industry code must appear in a **24-code digital-technology list** taken from the SEC's own classification.
- The deal must have an announcement date.
- **No minimum deal size.** Deal value is missing for about 70% of the database, so a size filter would silently discard cases on a data-quality accident rather than on substance.

Pool construction	Deals
Starting deal database (Thomson/SDC)	26,369
Removed: acquirer not resolvable on SEC	–21,651
Removed: target outside the technology screen	–3,658
Candidate pool	**1,060**

3.3 Stage 2 — Probe the SEC before downloading anything

This is the step that makes the method affordable.

- For each of the 1,060 deals, read the acquirer's **filing index** and ask a narrow question: *did this company file anything about this transaction?*
- The search window is the same one the retrieval code uses: **30 days before announcement to 30 days after closing**, or a year if no closing date is recorded.
- Outcomes are ranked, best evidence first:

Probe outcome	Meaning	Deals
`agreement_exhibit`	The merger or purchase agreement itself was filed	89
`merger_proxy`	A merger proxy or tender-offer statement was filed	81
`announcement_only`	Only a press release in the window	724
`no_transaction_filing`	Nothing filed in the window	166

- **170 deals were probe-positive.** For **112** of them, the acquirer's own filing **names the target**, which corroborates both the transaction and the company match from the filer's own document rather than from name similarity.

Two errors found and fixed at this stage

- **The window was initially too short.** A 60-day window classified Take-Two/Zynga and Intuit/Mailchimp as press-release-only, although both produced hundreds of passages in the pilot. Merger proxies and tender offers are filed **months** after announcement; Take-Two's landed at **day 133**. The probe now uses the retrieval window, so it predicts what retrieval will actually find.

- **Exhibit types were read from the wrong field.** The filing directory's JSON `type` field holds the **icon filename**, not the SEC exhibit type. Every agreement lookup silently failed. Exhibit types now come from the filing detail page.

3.4 Stage 3 — Retrieve

- We queued **400 deals**: all 170 probe-positive, plus the 230 press-release-only deals with the most filings in their window.
- **Why include the "weaker" group:** Microsoft/Nuance is press-release-only by this test and produced 231 passages in the pilot. Excluding them on principle would have discarded real evidence.
- That judgement was correct: **32 of the final 133 deals** came from the press-release-only group.
- Retrieval result: **35,296 documents, 96,936 pattern matches, zero failures.**

3.5 Stage 4 — Build the passage corpus

- A **passage** is one block of text from a transaction-linked filing that mentions employees, pay, benefits, equity, retention, severance, or employment terms.
- Documents unrelated to the deal are excluded even when the buyer filed them in the window. This matters: JPMorgan filed 2,379 documents in its window, of which almost none concerned the deal.
- Exact duplicates collapse to one row. Near-identical legal boilerplate is grouped into **provision families**, so one repeated clause cannot dominate the model.

Corpus construction	Count
Documents parsed	6,156
Documents kept as transaction-linked	2,862
Documents excluded as unrelated to the deal	3,294
Candidate passages screened	42,235
Passages included	**16,079**
Passages excluded	26,156
Provision families	34,295

What the screen throws away, and why

Reason for exclusion	Passages
Mentions a generic term with no people context	8,774
Navigation or index fragment	6,855
A bare heading such as "Employees" with no content	4,386
Accounting or financial context	2,466
Generic definition of "Representative"	898
Privacy or intellectual-property context, not employees	865

- The screen rejects more than it keeps, deliberately. A table of contents listing the word "Employees" is not evidence about employees.

Where the surviving text comes from

Filing type	Passages
EX-2.1 (the merger or purchase agreement)	4,821
424B3 (prospectus)	3,475
EX-10.1 (compensation exhibits)	1,944
S-4 and S-4/A (registration statements)	2,646
EX-99.1 (press releases)	678

3.6 Stage 5 — Freeze the sample with a yield gate

- A deal enters the model only with **at least 10 passages from at least 2 separate documents.**
- The threshold was set before the results were seen. Its purpose is to stop a deal contributing a single repeated clause.
- Deals that fail are **kept in the record**, not deleted.

Outcome for the 400 retrieved deals	Deals
Met the gate — the study sample	**133**
Below the gate	102
Retrieved, but zero employee passages	165

- **A zero-passage deal is a fact about disclosure practice, not about the company.** It does not mean the transaction had no employee arrangements. It means the public filings do not discuss them.

3.7 Stage 6 — The unsupervised model

The design question was: *what recurring themes exist in this text, without telling the model what to look for?*

Setting	Value	Reason
Features	Word and two-word phrase frequencies, weighted by rarity	Standard, interpretable, no black box
Method	Non-negative matrix factorisation	Produces additive, readable components
Number of themes tested	3 to 7	The stability tests choose, not the author
Random seed	20260823	Fixed, so the run is reproducible
Fit sample size	1,500 passages	About 11 per deal at this sample size
Balancing	By source-document family	**Prespecified before the run**

- The model is given **no category names**. It is never told that "equity" or "severance" exist.

- Theme descriptions in Section 5 were written **after** fitting, by reading each theme's top words and highest-weighted passages. They are the author's reading, not labels the model was taught.

On the balancing choice

- The 10-deal pilot showed that balancing by document family scored better than balancing by deal.
- That setting was therefore **fixed in writing before this run**, precisely so it could not be chosen after seeing which option flattered the results.
- The other two settings were run afterwards as sensitivity checks only.

3.8 Stage 7 — How the model is tested

- **Leave-one-deal-out:** remove each deal, refit, and check whether the theme reappears. Threshold set in advance: **0.80**.
- **Bootstrap:** 100 fixed-seed resamples.
- **A rival method:** an entirely different clustering algorithm run on the same text, compared for agreement. Threshold: **0.20**.
- **Sensitivity:** the whole model refit under all three balancing settings.

4. The sample

- **133 completed technology acquisitions**, announced 2020 to 2022.
- **13,817 employee passages.**

Property	Value
Announced 2020 / 2021 / 2022	35 / 66 / 32
Largest single deal's share of the text	**5.8%**
Deal value disclosed	107 of 133
Median disclosed value	$312 million
Largest deal	$43.5 billion (S&P Global / IHS Markit)
Target status: private / subsidiary / public / not stated	70 / 20 / 14 / 29
Source: agreement / proxy / press release	66 / 35 / 32

- The concentration figure is the important one. In the failed approach, one deal was **44%** of the text. Here the largest is **5.8%**, so **no single transaction drives the result.**

The ten largest contributors

Acquirer	Target	Passages
FiscalNote Holdings	Aicel Technologies	797
CF Acquisition Corp VI	Rumble	708
ADTRAN	ADVA Optical Networking	534

Acquirer	Target	Passages
System1	Protected.Net Group	492
Ginkgo Bioworks	Baktus (epidemiological unit)	476
Take-Two Interactive	Zynga	418
Advanced Micro Devices	Xilinx	403
Entegris	CMC Materials	314
Advent Technologies	UltraCell	308
Teledyne Technologies	FLIR Systems	287

5. Results

The model selected **three themes**. Each is described below with the words that define it, how much text it accounts for, and how well it survived testing.

5.1 Theme 1 — Executive and officer language

- **Defining words:** executive, officer, chief, employment, executive officer, chief executive, directors, employees, board, officers
- **Passages: 5,741.** Provision families: 4,947. Present in all 133 deals. Mean share of a deal's text: 35.5%.
- **Coherence: 0.258. Leave-one-deal-out recovery: 0.864.**

Reading (provisional). References to executives, officers, directors and the board. This theme **mixes two different things**: contractual executive-employment and change-in-control terms, and the officer or director language that appears in proxies and announcement press releases, including quoted remarks by a chief executive. Its coherence is the lowest of the three, which is what a mixed component looks like. It should not be read as a single kind of employee provision.

5.2 Theme 2 — Benefit plans and retirement law

- **Defining words:** plan, benefit, ERISA, employee, benefit plan, code, pension, employee benefit, plans, benefit plans
- **Passages: 5,762.** Provision families: 5,016. Mean share of a deal's text: 42.7%.
- **Coherence: 0.351. Leave-one-deal-out recovery: 0.889.**

Reading (provisional). Definitions and covenants about employee benefit plans: plan schedules, retirement-law representations, pension and welfare arrangements, and undertakings about what benefits continue after closing. This is the most self-consistent contractual theme, and the largest by share of the average deal's text.

5.3 Theme 3 — Equity awards at the closing moment

- **Defining words:** stock, shares, common, common stock, restricted, restricted stock, units, options, merger, outstanding
- **Passages: 4,576.** Provision families: 3,725. Mean share of a deal's text: 21.9%.

- **Coherence: 0.432. Leave-one-deal-out recovery: 0.983.**

Reading (provisional). What happens to employees' stock, options, restricted stock and restricted stock units at the moment the transaction closes: conversion, assumption, vesting treatment, and cash-out of outstanding awards. It is the most internally coherent and the most stable theme of the three.

5.4 How well the themes held up

Theme	Passages	Coherence	Leave-one-deal-out recovery	Passes the 0.80 bar?
Executive and officer language	5,741	0.258	0.864	Yes
Benefit plans and retirement law	5,762	0.351	0.889	Yes
Equity awards at closing	4,576	0.432	0.983	Yes

- **71 automated checks passed. 0 failed. 1 warning.**

The comparison that matters most

Model	Deals	Overall recovery	Verdict
The 10-deal pilot	10	**0.630**	**Failed** the 0.80 bar
This study	133	**0.864–0.983**	**Passed** on every theme

- Same method, same thresholds, more deals. **The pilot's failure was a sample-size problem, not a method problem.** This is the single most useful methodological result of the project.

5.5 Sensitivity: does the answer depend on how we built it?

Balancing setting	Themes found	Recovery per theme	Leading words
Source-family (**primary, prespecified**)	3	0.86, 0.89, 0.98	executive; plan; stock
By deal	3	0.95, 0.95, 0.95	employment; stock; plan
Unbalanced	3	0.90, 0.94, 0.98	executive; plan; stock

- **All three settings return the same three themes**, in different proportions, and every component clears the 0.80 bar.
- The themes are a property of the text, **not an artefact of the analyst's configuration.**

5.6 The one warning, stated plainly

- A **completely different clustering algorithm** was run over the same passages and its grouping was compared with the model's.
- Agreement score: **0.034**, against a threshold of **0.20**.
- **What this means:** the three themes are each individually stable — they reappear when any single deal is removed — but **a different algorithm would divide the same text differently.**

- **How to read the themes because of it:** as *recurring patterns of language*, not as *the one correct taxonomy* of employee provisions.

5.7 The AI subgroup

- Applied **after** selection, to deals already known to have employee disclosure.

Label	Deals
Filings describe the target in explicit AI terms	**17**
Weaker or adjacent AI language	0
No AI language near the target's name	116
Deals using language about a team "joining" the buyer	45
Deals with explicit acqui-hire language	3

- Named examples with the wording their own filings use: Microsoft / Nuance, Hewlett Packard Enterprise / Silver Peak, DocuSign / Seal Software ("artificial intelligence", "natural language"), Intercontinental Exchange / Ellie Mae, Progress Software / Chef, Take-Two / Zynga ("machine learning"), SentinelOne / Attivo Networks ("AI-powered").
- **17 of 133 is itself a result.** Among transactions whose employee terms are public, only a minority are described in AI language at all.
- This is **technology M&A with an AI subgroup inside it**, not an AI-deal sample. Every label is machine-derived and pending human review.

5.8 Tone, as a secondary diagnostic only

- Counts protective and negative wording per hundred tokens, then subtracts the average for the same filing type, so deals are compared against **ordinary legal language** rather than against plain English.
- Highest protective-language residuals: Opendoor / Commeasure (+5.33), Vontier / DRB Systems (+1.83), KKR / Therapy Brands (+1.57).
- Recorded in the manifest as `secondary_diagnostic_corpus_not_validated`.
- **This measures drafting style. It is not evidence that any buyer treated people better.**

6. The separate deal-architecture layer

- A second, distinct layer codes **what kind of transaction** each of the ten original pilot deals was. This is **rule-based human coding, not unsupervised learning**, and the two are never mixed.
- Six attributes per deal: legal form, scope and control, IP treatment, business continuity, workforce movement, and explicit talent motive.
- A strict rule was introduced: **every asserted claim must quote the SEC document word for word.**

Treatment of the 70 register rows	Rows
Now quote the filing verbatim, each verified as an exact substring	**20**
Withdrawn to "unknown" because no sentence in the document states it	23
Already recorded as unknown	27

- **Withdrawing is the safe direction.** A nearby but unsupporting quote looks like evidence while proving nothing, which is worse than an honest paraphrase.
- **Example of a withdrawal:** Microsoft / Nuance's only reviewed document is a press release saying "acquisition". That cannot support a claim of *statutory merger*, so the legal form is now unknown. Its leadership-continuity claim survives, because the release states that **"Mark Benjamin will remain CEO of Nuance, reporting to Scott Guthrie."**
- All nine IP-treatment rows were withdrawn; the register itself recorded that no reviewed passage addresses IP.
- Resulting archetype suggestions: full acquisition 4, asset acquisition 1, asset acquisition with talent salience 1, unknown 5. All are **machine suggestions pending human review**, and the three human-decision columns remain blank.

7. What the results mean

7.1 The substantive reading

- **Employee terms in technology acquisitions cluster into a small number of recurring instruments.** Across 133 deals and 13,817 passages, three themes account for the text: benefit-plan continuity, equity-award treatment at closing, and executive-level employment and officer language.
- **Benefit continuity is the largest single category** by share of the average deal's text (42.7%). The most common thing a merger agreement says about ordinary employees concerns their health and retirement plans.
- **Equity treatment is the most standardised.** Its recovery of 0.983 and its high coherence say that acquirers describe option and share conversion in strikingly consistent language across very different deals.
- **Executive language is the least standardised.** It mixes contract terms with announcement rhetoric, which is a finding about *where* companies talk about people: senior individuals appear in press releases and proxies, ordinary employees appear in the agreement's covenants.
- **Explicit talent framing is rare in the legal record.** Only 3 of 133 deals use explicit acqui-hire language, and 45 use "joining" language. The talent story that dominates press coverage is largely absent from binding documents.

7.2 The methodological reading

- **Selecting research samples by subject matter fails when the evidence is disclosure-dependent.** This is the transferable lesson. The AI-first screen was not badly implemented; it was asking for evidence that structurally does not exist.

- **Sample size, not method, was the binding constraint.** The identical pipeline scored 0.630 on 10 deals and 0.864–0.983 on 133.
- **Cheap existence checks before expensive retrieval change what is feasible.** Probing 1,060 deals took about 40 minutes and turned a blind crawl into a targeted one.

7.3 What this does not mean

- It does **not** mean employees at these companies were retained, paid as promised, or satisfied.
- It does **not** mean a buyer with more protective language treated people better.
- It does **not** establish cause and effect in any direction.
- It does **not** describe acquisitions in general. See Section 9.

8. Reproducibility

- Every number in this report is read from a manifest or table produced by the pipeline. The report generator computes no statistics of its own, so the document cannot drift from what the code produced.
- The result tables are published in the repository at `data/published/disclosure_sample_133/` (410 KB, all 400 probed deals with the 133 modelled ones marked).
- Excluded from publication with the reason recorded: the 117 MB passage corpus, the 35,296 retrieved documents, the HTTP cache, and the licensed vendor archive.
- Selection rule `disclosure-first-v1`. Corpus hash `97ea426f6900a345…`.
- Software state: **269 tests passing**, lint and type checks clean.

```
tag-edgar screen-disclosure-pool data/derived/deal_catalog.csv
tag-edgar probe-disclosure data/derived/disclosure_pool/pool.csv
tag-edgar build-disclosure-queue data/derived/disclosure_probe/probe_results.csv
tag-edgar run-disclosure-sample data/derived/disclosure_review_queue.csv
bash scripts/run_disclosure_analysis.sh
```

9. Limitations

9.1 The sample is selected by disclosure

- It describes acquisitions whose **buyers file with the SEC and put the agreement on the record.**
- Deals by private, private-equity, and foreign buyers are largely absent — **21,651 of 26,369** were removed for exactly this reason.
- This is **a property of the public record, not a sampling error that weighting can repair.** No amount of statistical adjustment recovers documents that were never filed.

9.2 The corpus relevance audit was not run

- The intended check: a person reads **150 sampled passages** (75 kept, 75 rejected) blind, and confirms the screen kept the right ones. The bar is 90% relevance among kept passages and under 5% missed content among rejected ones.

- It was **skipped for time** in this cycle, by direction.
- **Why this matters concretely:** the same audit run on an earlier corpus scored **72% against a 90% threshold**, with 5.33% missed content. So the risk that a meaningful share of "employee" passages are not really about employee treatment is **real and already measured**, not hypothetical.
- Consequence, enforced by the software: every downstream artifact is labelled provisional and the report verdict is withheld. The blinded packet is prepared and can be labelled later, upgrading the results **without re-running any model**.

9.3 These are documents, not outcomes

- A retention bonus is a contractual design. It is not proof anyone stayed.
- A promise of benefit continuity is a promise, not an observation.

9.4 The theme descriptions are provisional

- They were written after seeing the model's output and have **not** been scored by two independent reviewers. The blinded review packet exists and is unlabelled.

9.5 Theme 1 is a mixed component

- Its low coherence and its mixture of contract text with press-release quotes mean it should not be treated as a single provision type.

9.6 Company matching is machine-confirmed

- For 112 deals the acquirer's own filing names the target, which is strong corroboration. **No person has checked each pairing.**

9.7 The architecture layer covers ten deals, not 133

- Only the ten original pilot deals have reviewed transaction-structure evidence, and even there the human review columns are blank.

10. What would strengthen this next

Ordered by value gained per hour spent.

1. **Label the 150-row relevance packet.** About one hour for two people. This single act moves every result from provisional to validated, or tells us honestly that the screen needs repair.

2. **Complete the two-reviewer theme review.** 30 items, roughly 20 minutes each. It converts the theme names from one person's reading into a measured agreement statistic.

3. **Separate announcements from agreements and refit.** Theme 1 mixes the two. Splitting them would test whether executive language is really one theme or two.

4. **Extend the years.** The database covers 1980–2022 and we used 2020–2022. Widening would enlarge the AI subgroup beyond 17 deals.

5. **Decide whether the architecture layer scales.** Either extend rule-based coding to all 133 deals with machine-only fields, or keep it as a deliberate ten-deal deep dive.

11. Questions for Dr. Singh

- Is a **disclosure-selected sample** acceptable for this research question, given that private and foreign buyers can never be included?
- Should the 150-passage human audit be completed before anything is presented as a finding?
- Theme 1 mixes contractual executive terms with press-release rhetoric. **Should announcements be excluded** from the corpus and the model refit?
- Is **17 AI-labelled deals** a usable subgroup, or should the year range widen first?
- Should the deal-architecture layer be **rebuilt at 133 deals**, or remain a ten-deal deep verification alongside the large-sample text analysis?